when it comes to construction of this antenna, it is difficult to find its proper construction details. Further, when you have enough information to start building the antenna, the availability of material locally and at affordable cost is a big problem. If you decide to use the material available in your junk-yard, you cannot be sure about performance of this antenna. So, better to be cautious. The QFH antenna is shown in Fig. 6.

Fig. 10: Vee-dipole (left) and turnstile (right) antenna mounted on the poles

Fig. 11: Typical BNC type connectors: BNC-to-SMA adaptor (left), BNC female socket (centre), and BNC male plug/header (right) used for the cable assembly

Fig. 12: Receiver RSP1A from SDRPlay (left) and RTL-SDR dongle V3 (right); cable assembly and BNC-to-SMA adaptor are in the fore

TABLE 1
PARTS LIST

Item	Needed
6mm aluminium tube	1x1.2 metre
General-purpose PCB/plastic box (50mm x 75mm x 25mm)	1
BNC female header (optional) panel type or PCB type	1
BNC male plug (cable type)	1
BNC female plug (cable type)	1
SMA male-to-BNC male adaptor for receiver	1
SDR receiver (RTL-SDR or RSP1A)	1
RG6 or other 75-ohm coax cable	10-15 metre
M3 x 15mm screws	4
M3 hex nuts	4
M3 x 20mm self-tapping screws	2
1.27cm PVC plumbing pipes (3 to 4.5 metre feet heigh)	1-2
Miscellaneous hardware/fitting items	

The turnstile antenna. The next alternative is the turnstile antenna. It utilises two linear dipoles placed at 90 degrees or at right angle to each other, with one dipole connected to the necessary 90-degree phasing cable (quadrature connection). This antenna is simpler to construct as it uses only two linear dipoles. However, it has poor near-the-horizon performance, which is required to pick up satellite signals from the horizon or low elevation angles.

The author tried one such combination and found that it offers higher signal gain, hence higher image quality, but only in the ±30 degrees field-of-view (FoV) about the vertical, that is, 120 degrees in total and only when the satellite is overhead and has lesser gain. Almost no image data could be seen near the horizon angles. The turnstile antenna is shown in Fig. 7.

The Vee-dipole antenna. The simplest form of the antenna is the V (or Vee) dipole, which is just an extension of the single linear dipole. In fact, the two quarter-wave sections of a linear dipole are placed or mounted at an angle of 120 degrees as against 180 degrees for the straight linear dipole. You can start with this simple but effective antenna. The author had a linear dipole primarily optimised and tested for the 2m (144MHz) amateur radio band centred at 145MHz.

So, you can begin by constructing this V-dipole. The 137MHz band and the satellite frequencies range from 137.1MHz all the way to 137.9125MHz. Let's assume 137.5MHz as the central frequency. The free-space wavelength is 2.1818m at this frequency, the half-wavelength is 1.09m, and the quarter-wavelength is 0.54545m.

You shall require a 6mm (approximately ¼-inch) diameter aluminium tube of 1.2m length, which can be obtained from a local hardware shop at very low cost. Cut the tube into two 60cm long identical sections.